

Much of the debate over the deskilling of work has assumed that skill is simply technically derived. Here it is argued that this is a partial view and that the categories for evaluating skill definitions are gender-biased. An adequate understanding of gender inequality at work must take account of both the ideological aspects and the material components of skill. The centrality of men's control of technology to their power in the workplace is discussed. The article concludes that social relations are expressed in and shape technologies themselves and that patriarchal relations are integral to this process.

Patriarchy, Technology, and Conceptions of Skill

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The debate on the impact of technology on the labor process has come a long way since the publication of Braverman's (1974) *Labor and Monopoly Capital*. It has been convincingly argued that any simple tendency toward deskilling is belied by the complexity of reality. There has been relatively less analysis of the key concepts in the debate—skill and technology. By arguing that the categories for evaluating skill definitions are themselves gender-biased, the limitations of a purely technical definition of skill will become apparent.

Recent feminist work has emphasized that distinctions of skill between women's and men's work frequently bears little relation to the actual amount of training or ability required for them. However this formulation, which stresses the social/political construction of skill, is now in danger of losing altogether any element of the material realities of skill. And a key term, technology, itself remains unexamined, as in standard technological determinist arguments.

Writers such as Noble (1984) have examined the way in which capitalist social relations affect technological design: that technology itself must be seen as "frozen social relations." My argument here is that male power is also embodied in the design of technology. Technology is made by men to make

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full use of their physical advantage and technical competence. The workplace culture based on technical skill, which expresses and consolidates relations among men, is an important factor in explaining the continuing exclusion of women from skilled work. Perhaps technology also represents "frozen gender relations."

BRAVERMAN AND DESKILLING

Debates on work and technology have been influenced by what is known as "the labor process perspective" or the deskilling theme, initiated by Braverman. This theory has been part of a more general critique of technological determinism, which saw technology as constituting an autonomous force determining the organization of work. The basic argument of the labor process literature which developed was that within capitalism, a major factor affecting the technology of production is the class relations of production. According to this view, capitalism requires the continuous application of new technology to the fragmentation and cheapening of labor, resulting in deskilling.

While women have always worked in large numbers, it is over the past 3 decades that they have come to comprise nearly half of the labor force in advanced industrial economies. Although Braverman did provide a framework for looking at women workers and the effects of technology on their skills, the existing sexual division of labor is taken for granted. Even so, many of the protagonists in the debate have been oblivious to gender issues, implicitly concerned only with those sectors of production in which male workers predominate. The proliferation of studies on machine tools is a case in point. Since the mid-1970s, however, feminist researchers and activists have insisted that the effects of automation on women's employment be addressed.

There has been a rapid growth of research on women and technology since 1976, when Cowan first argued that the female experience of technological change was significantly different from the male experience. Zimmerman's *The Technological Woman* and Rothschild's *Machina ex Dea*, both published in 1983, were concerned to recover women's contributions to technological innovation as well as to examine the impact of technological change on women's work. Recent collections, such as those edited by Hartmann, Kraut, and Tilly (1986-1987) and Wright (1987), broadly examined the same issues.

Although some of these authors believe that much of the routine work that women do will disappear and be replaced with more skilled work, much more

common among feminist writers is a pessimistic view of the impact of microelectronic technology on women's work, often expressed in a strongly antitechnology stance.

The introduction of information technologies is seen as part of the general process of deskilling, the purpose of which is to increase productivity and profit, in this case by deskilling typists and incorporating the monitoring of work into the machinery itself. It has also been suggested that computerization of office work would lead to a huge reduction in the number of office jobs and the emergence of the "paperless office." Labor process authors characterize the office as a white-collar replica of the assembly line, with office work divided into many subtasks, each performed by a specialized worker who loses both contact with the total product and variety in the tasks performed. Word processors are seen as a threat to typists' skills which are being built into the new machines. Secretarial work for those few who remain will be increasingly fragmented into routine and standardized tasks subject to the control of the machine. The result is that workers have less and less control over the production process. For example, two well-known studies by Feldberg and Glenn (1983) and Crompton and Jones (1984) documented the disproportionately negative effects of automation on the job content and autonomy of women's jobs. They argued that job loss is concentrated among low-skilled female clericals and that women are increasingly confined to poorly paid jobs with few opportunities for promotion. Indeed, much of this feminist research on women workers simply replicated Braverman's approach in developing a mechanistic analysis of deskilling.

Neither a paperless office nor a vast pool of unemployed secretaries has materialized. There has now been a considerable critical response to this literature, particularly questioning the existence of any simple tendency toward either the deskilling or the upgrading of labor. Empirical studies of technological change have repeatedly shown that reality is more complex and does not develop in a unilinear direction (Appelbaum, 1987; Baran, 1987; Carter, 1987; Webster, 1989). Indeed, with respect to the skill levels required for given jobs, opposing tendencies of increased complexity and of greater simplification and standardization have coexisted. For example, in relation to office automation, many studies have assumed that the machinery itself had some inherent logic which caused its impact to be uniform across the range of office jobs (Barker & Downing, 1985; West, 1982). Detailed case studies have shown, however, that identical equipment, in this case word processors, may in fact have very different effects on work experience.

Webster's (1989) British study, comparing the effects of new technology on typists and secretaries, found that automation operated within and rein-

forced preexisting differences in the patterns of work. The rationalization and fragmentation of clerical work had, in fact, long predated the advent of computer technology, and its introduction reinforced the tendency for typists to perform repetitive, standardized tasks. At the same time, however, word processors reduced the burden of routine work for the secretaries, enabling them to continue to do a variety of relatively responsible tasks. Thus the introduction of word processors exacerbated preexisting divisions between the two groups of women office workers, enhancing the position of some secretaries but not that of typists.

In addition, it is now widely recognized that a major problem with Braverman's approach was his deliberate exclusion of working-class consciousness and organization in the labor market. This resulted in his neglecting how labor can influence the production process through resistance and compliance. The form of jobs is always the result of a social process involving choice and often negotiation between management and labor. At times, this is newly established and at others, it rests more on tradition and past practice. Braverman did not fully recognize the influence of social processes which mediate between the technology and levels of skill on its introduction. The threat of deskilling may precisely "induce defensive actions on the part of the workers to stratify the labour force, control entry to occupations and maintain skill status long after these skill divisions have become irrelevant" (Rubery, 1978, p. 27).

Numerous studies of labor processes have included the question of worker resistance and struggle in shaping the labor process. This focus led to studies on the struggle between capital and labor at the point of production, particularly labor organized in craft unions. The deskilling of craft workers was, and largely still is, the dominant reference point for these analyses. However, a major problem with these studies was the assumption that the social relations within which technology developed could be understood simply in terms of relations between worker and capitalist. This underestimated the continuing significance of divisions within the working class.

There has since been increasing recognition of the ways in which conflicting interests over the outcome of technical change are likely to emerge within the work force. Feminist writers in particular have exposed the deficiencies in an analysis of technological change exclusively focused on class conflict. The standard historical accounts of craft unionism examined the role of technical skills in securing job control, that is, as a weapon in class conflict. Its role as a weapon in patriarchal struggles at work is ignored. It is now well established within the feminist literature that as exclusively male preserves, craft unions have played an active part in creating and sustaining women's

subordinate position in the work force. The identification of men with skill has been central to male dominance in the workplace. We need to understand changes in the organization of work not only as the outcome of continuous conflict between capital and labor, but as conflict between workers themselves, especially between male and female workers.

NEW TECHNOLOGY AND GENDER RELATIONS

The most interesting work on patriarchy, skill, and technology has taken up this issue of divisions between workers and focused on the relationship between men and women in the workplace, the implications for the construction of jobs and the sex-typing of occupations. This has been part of the growing recognition of the limitations of sociological accounts which analyze women's position in the labor force primarily with reference to the domestic division of labor.

This is not to deny that women's disadvantaged position in the labor market is in large part due to their greater responsibility for dependent care and household tasks, as I explored elsewhere (Wajcman, 1983). However, gender relations are embodied in the sphere of production, as well as in the sphere of reproduction. Thus the gender stereotyping of jobs is not just a reflection of women's traditional role within the family; it is also formed and reproduced by the patriarchal relations of paid work.

Some commentators have presumed that with technological developments, such as the elimination of much heavy physical work by mechanization, the boundaries between women's and men's work would gradually disappear. From a different perspective, the labor process literature presumed that women would become fully integrated into the labor force as technology led to its increasing homogenization (Liff, 1986, p. 75).

However, the gender stereotyping of jobs has remained remarkably stable even when the nature of work and the skills required to perform it have been radically transformed. The broad nature of gender divisions in the labor market is well established: Men and women are segregated into different occupations, and this segregation is particularly marked within individual workplaces. Women are, on average, paid about three quarters of men's hourly earnings. What requires explanation is the contrast between the flexibility of the form taken by occupational segregation by sex and the persistence of its existence.

It has been more common for women to enter new jobs requiring new skills than to break into traditional male preserves. However, even the

allocation of these completely new jobs, where no gendered custom and practice has been established, is a fundamentally gendered process. In new "high tech" jobs, such as programming, women tend to be segregated into positions at the bottom of the occupational hierarchy (Kraft & Dubnoff, 1986). Game and Pringle (1983) and Cockburn (1985) presented case studies of employment where technical developments have substantially altered the skill and task range of jobs and yet the distinctions between "men's" work and "women's" remain, with men still monopolizing the skilled, technical jobs. In a sense, then, much of the recent feminist work has addressed the issue of why there has been so little change in the degree of sex segregation of the labor market and so little conflict over the continuing rigid sex-typing of occupations.

What role does technology play in the construction and reproduction of these gender relations and in their potential transformation? New technologies do disrupt established patterns of sex-typing and thereby open up opportunities for changing the sexual division of labor. As technologies develop and displace each other, there is a disturbance among the technically skilled strata. Some gain and some lose. Many male craft skills have been quite purposively made redundant by new technology which has radically transformed the nature of the work (Gill, 1985; Wilkinson, 1983). But technology is not an independent force; the way in which it affects the nature of work is conditioned by existing relationships. There are conflicts and negotiations over technological change and the opportunities for changing the sexual division of labor to women's advantage are often foreclosed by male power.

Women lose out in these struggles as powerful groups defend their old skills or monopolize new ones. Craft workers, who have been seen as the defenders of working-class interests in struggles over technical change, partly derive their strength from their past exclusionary practices (Penn, 1989). The crystallization of craft control has often been at the expense of less skilled or less well-organized sections of the work force, and this has, in many cases, involved the exclusion of women.

The entry of women into industrial work in Great Britain, America, and Australia during World War I and, especially World War II was followed by an equally deliberate process of their expulsion from that work once the immediate crisis had passed. Milkman (1987), for example, described the massive entry of women into war industries like auto and electrical manufacturing during World War II. Rather than explaining the postwar purge of women from men's jobs as simply the result of the appeal of domesticity, Milkman argued that it was the combined efforts of management and unions

that resulted in their removal. According to her, most women workers actually wanted to retain their jobs. Thus the gross underrepresentation of women in engineering and allied industries and the lack of confidence often felt by women faced with technology are evidence of a deeper problem. Official plans to rectify the underrepresentation of women in engineering often proceed as if the problem were simply to improve women's self-confidence. But male dominance of technology has largely been secured by the active exclusion of women from areas of technological work.

Printing and newspaper publishing in particular is an industry with craft traditions of labor process control. Recent developments, particularly in electronic typesetting technologies which have the potential to undermine those traditions, have been resisted by printing workers. Strikes and lockouts throughout the 1970s and early 1980s characterized attempts to introduce the new technology in the United States, Great Britain, and Australia. The printing industry in Britain provides a contemporary illustration of the sexual politics involved in such struggles over technology.

The violent dispute at the new technology newspaper plant at Wapping in London during 1986 was the final phase in a long history of management attempts to wrest control of the labor process from the Fleet Street print unions. Computerized photocomposition systems had been available in the United States printing industry since about 1970. This technology enabled journalists and advertising personnel to enter copy directly into a computer. The introduction of this new technology represented an attack on the compositors' control over their work, as it meant that their traditional manual skills would become technically redundant.

The restrictive practices of craft labor and the degree of chapel control had, not surprisingly, inhibited the introduction of this kind of technological innovation into the newspaper industry in London's Fleet Street. Wages of these craftsmen have been very high in the postwar period. Those of compositors, who prepared the type in hot metal, were the highest of all. In her book on the history of typesetting technology in Britain, Cockburn (1983) described this archetypal group of skilled male workers as they were being radically undermined by cold electronic composition. This is an area of work from which women have been traditionally excluded. For employers, the new technology represented, among other things, an opportunity to replace the men with cheaper women workers.¹ Over the past decade, the compositors have fought to defend their position by having sole rights to use the computer typesetting equipment—to retain keyboard work. To varying degrees, they managed to maintain their craft control, even though their craft was technologically redundant. However, their strategy of resistance has entailed the

exclusion of unskilled women from the trade. It should be noted that this exclusionary strategy has also involved racial and religious prejudice. In fact, as Cockburn noted, the skilled in printing have a higher proportion of White Anglo-Saxon Protestants among them than the semi- and unskilled.

Having pointed out the way in which organized male workers have used their monopoly of technology to maintain power over women in the workplace, it needs to be said that this is not a once and for all achievement. Male dominance over machinery is constantly under threat by women's direct efforts to undermine it and by actions of employers seeking to undermine skilled male workers and cheapen their labor costs.

Under some conditions, skilled men do lose out and women enter previously male jobs. The process of feminization is often part of technological change. In such cases, women rarely perform exactly the same tasks, under the same conditions, as the men formerly performed. Inherent in this process of technological change is the transformation of jobs. However—and this is the crucial point—the introduction of female labor is usually accompanied by a downgrading of the skill content of the work and a consequent fall in pay for the job.

SEX, SKILL, AND TECHNICAL COMPETENCE

This immediately raises the question of the meaning of skill itself. It is often said that women are low paid because they are unskilled, and certainly women's work tends to be classified as unskilled or semiskilled in official classifications. But the crucial question is how definitions of skill are established. To take a simple example, women who assemble digital watches and pocket calculators require considerable manual dexterity, the capacity for sustained attention to detail, and excellent hand-eye coordination. Yet these capacities are not defined as "skills." Nurses provide another example of an occupation that requires a great deal of training and ability as well as technical knowledge. However nursing is not thought of as a technical job because it is women's work and therefore undervalued. The work of women is often deemed inferior simply because it is women who do it.

How has it come about that women have failed to achieve recognition of the skills required by their work? Although it is the case that women workers have generally been refused access to training in traditionally masculine areas of work, the basis for distinctions of skill in women's and men's work is not a simple technical matter. The classification of women's jobs as unskilled and men's jobs as skilled frequently bears little relation to the actual

amount of training or ability required for them. Skill definitions are saturated with gender bias. Women workers carry their status as subordinate individuals into the workplace, and this status comes to define the value of their work. As Phillips and Taylor (1980) put it, "far from being an objective economic fact, skill is often an ideological category imposed on certain types of work by virtue of the sex and power of the workers who perform it" (p. 82).

The Australian clothing industry provides a clear illustration of the way that skill definitions are themselves gender-biased (O'Donnell, 1984).² Throughout this century, machining in the clothing trade has been done by both women and men. But where it was done by men, the work was classified as skilled whereas it was classified as unskilled when done by women. Today, 70% of the production employees in the industry are "unskilled" machinists, and they are invariably female. The second largest group of workers is "skilled" cutters, who are mostly male. This is reflected in their differential wage rates: Machinists earn significantly less, and their system of payment is known as piece rates or payment by results.

O'Donnell's research demonstrates that an explanation for the difference in wage rates has little to do with skill and more to do with the place of the cutter in the production process and the actions of the union. Historically, the men's suiting area has been the place of the most active unionists, an area of predominantly male employment. The position of cutter is a powerful one, in that the entire production process can be held up by a small group who stop cutting cloth. Management has exploited this division between cutters and machinists in order to keep the wages bill down. The numerically small, more militant cutters have created a labor market shelter for themselves, while the majority of employees, the machinists, receive the lowest possible wage.

These clothing workers are in a weak bargaining position not because of their lack of skill but because of the large pool of married migrant women whose social situation forces them to compete in this area of work. It is not possible for anyone to sit down at a sewing machine and sew a garment without previous experience. To be a competent machinist demands considerable knowledge and experience with the machine. This technical skill is also devalued and underpaid because it is seen as women's supposed natural aptitude.

Definitions of skill, then, can have more to do with ideological and social constructions than with technical competencies which are possessed by men and not by women. It is a question of workers' collective efforts to protect and secure their conditions of employment by retaining skill designations for

their own work and defending that skill to the exclusion of outsiders. These efforts, predominantly by and on behalf of the male working class, have been directed against employers who have regularly tried to find ways of substituting cheaper workers for expensive skilled labor.

But men's resistance has also operated against women's interests. Defending skill, preventing "dilution," has almost always meant blocking women's access to an occupation. Moreover, employers' own perceptions of the suitability of women for particular types of work must, in part, be responsible for the craft workers' success in excluding women from skilled work (Liff, 1986). Otherwise, one would expect the sexual division of labor to be a much more contested area both for management and unions than it is. Skilled status has thus been traditionally identified with masculinity and as work that women do not do, while women's skills have been defined as nontechnical and hence undervalued.

To avoid conceptual confusion, it is worth noting that the feminist analysis of skill as ideological can be overstated. This argument has been extremely influential in countering the conventional view that skill is an attribute of a job which is simply technically derived. However, in itself, it too is only a partial conception of skill. To fully understand gender inequality at work we need to recognize *both* the ideological and the real material aspects of skill. While it is the case that the gender bias of skill definitions is indicative of women's relative lack of power, there are also important connections between women's relative lack of power and their lack of technological skills. Men's dominance over machinery has been and continues to be a tangible basis for their power in the workplace. It is important to stress that technology includes not just things themselves but the physical and mental know-how to make use of these things. Know-how is a resource that gives those who possess it a degree of actual or potential power and, as seen in the foregoing cases, this know-how has been central to the class politics of technological work. It is also central to the sexual politics of technological work, as technical competence is a key source of men's power over women—of the capacity, for example, to command higher incomes and scarce jobs.

How can we begin to understand the enduring force of this identification between technical skills and masculinity without making the mistake of treating technology as inherently masculine? We can start by taking seriously the requirement to understand the masculinity of technology as a social product. Men's affinity with technology is then seen as integral to the constitution of male gender identity.

Once we recognize that gender construction is an ongoing ideological and cultural process with a long history, then the focus shifts to analyzing the

social practices involved. The way in which the present technical culture expresses and consolidates relations among men becomes an important factor in explaining the continuing exclusion of women.

This type of analysis stresses the importance of the cultural aspects of gender relations and shows the way that gender is an integral part of people's experience in the workplace. This is illustrated in Cockburn's (1983) study of compositors cited earlier, where she ascribed the centrality of the craft workers' ease with technology to their masculine identity. The industrial power of craftsmen derived from their knowledge and competence with machines. The control over this type of industrial technology has traditionally been the province of men, and women workers have been excluded from these technical skills. The technical change from Linotype to electronic photocomposition, however, makes the compositors feel emasculated. Because the work of composing now resembles typing and involves working with paper instead of metal, that is a shift from manual work to the office. The compositors no longer consider it as real work. Traditional craft culture was associated with hot metal, dirt, and physical work, and the elimination of this not only diminishes their control over the work but represents a threat to their masculinity.

Clearly, however, the appropriation of technical skills plays an important part in the reproduction of inequality among men as well as between men and women. Men do not have power over women in the same sense as employers do over workers. In looking at the relations of work, one is inevitably looking at class relations. The male culture of craft know-how is the culture of an exploited group. Male employees themselves vary considerably in their capacities to control and benefit from technological innovations. It is important to remember that technology is also used by some men to dominate others.

THE SOCIAL SHAPING OF TECHNOLOGY: INDUSTRIAL CONFLICT AND TECHNICAL INNOVATION

Thus far, I have examined the relationship between technology and the construction of skill. However, I have not raised the question of what has shaped the particular technological developments in the first place (MacKenzie & Wajcman, 1985). New technology typically emerges from modifications to and combinations of existing technology. However, this is not the only force shaping technology. Industrial innovation is a product of a historically specific activity carried out in the interests of particular social groups and

against the interests of others. If technology is designed with job stereotypes in mind, then it is hardly surprising that sex segregation is being further incorporated into the workplace. The remainder of this article explores that ways in which the sex of the work force and gender relations in the workplace themselves profoundly affect the direction of technological innovation.

Before doing this, however, it is pertinent to look at the exemplary study of the way in which capitalist social relations shape the design of technology. The central argument of Noble's (1984) *Forces of Production* is that patterns of power and cultural values shape the actual processes of technological development. Noble argued that the concepts of "economic viability" and "technical viability," often used to explain technological change, are inherently political. By way of a detailed reconstruction of a lost alternative to numerical control and by examination of variant forms of numerical control that have also vanished, Noble showed that automation did not have to proceed in the way that it did. Rather, the form of automation was the result of deliberate selection.

Noble suggested that it was at least partly the social relations of production that tipped the balance in the choice of technology. Machining was automated, using the technique of numerical control. But there was also a technique of automation called record playback. With record playback, the control of the machine remained with the machinist, whereas numerical control appeared to offer a means of dispensing with these well-organized skilled workers. Management, argued Noble, decided to take advantage of this possibility. Thus the technical choices made can be understood only by paying attention to the conflictual relations of production within which machining takes place.

It would be wrong, however, to assume that the goals of managers in preferring numerical control to record playback were necessarily realized. The introduction of numerical control on the shop floor did not simply shift control to management. It was met with fierce resistance from the work force. At the same time, management found that it needed to retain skilled machinists to operate the new machines effectively. Consequently, management was never able to gain complete control over production. In reality, machines do not run themselves, and, therefore, the tendency to deskilling is always contradictory.

Although it is class divisions which are central to Noble's analysis of the way by which industrial conflict shapes technological innovation, patriarchal relations are also of fundamental importance for any discussion of the development of technology. One of the ways that gender divisions interact with technological change is through the price of labor, in that women's wage

labor generally costs considerably less than men's. This may affect technological change in at least two ways. First, as we have seen earlier, employers may seek forms of technological change that enable them to replace expensive skilled male workers with low-paid, less unionized female workers. Second, because a new machine has to pay for itself in labor costs saved, technological change may be slower in industries where there is an abundant supply of low-cost women's labor.

There is also a more direct sense in which gender relations leave their imprint on technology. Recent feminist work has emphasised that distinctions of skill between women's and men's work have as much to do with job control and wage levels as with actual technique. However, as indicated earlier, this formulation understates the tangible and material basis of skill. Men selectively design tools and machinery to match their technical skills. Machinery is designed by men with men in mind. Industrial technology thus reflects male power as well as capitalist domination.

THE GENDERED RELATIONS OF TECHNOLOGY

The best examples of the gendering of technology come from Cockburn's (1983) history of typesetting, outlined earlier, which provides a detailed description of the technological evolution of the computerized photocomposition system. Like Noble, she showed that automation did not have to proceed in the way that it did. Rather, the form of automation was the result of deliberate selection. Cockburn suggested that the technical choices made can only be understood by looking closely at the conflictual relations of production, including gender relations which occupied a central place.

Computerized photocomposition technology has what is known as a QWERTY keyboard. This is not the standard keyboard incorporated into computers. However there was nothing inevitable about this. Electronic circuitry is perfectly capable of producing a Linotype layout on the new-style board. The lay of the Linotype keyboard differs greatly from QWERTY. Not only does it have 90 keys in contrast to 44, the relative position of the letters of the alphabet also differs from that of a typewriter and the keys are larger and spread further apart.

So, what politics lie behind the design and selection of this keyboard? In choosing to dispense with the Linotype layout, management was choosing a system that would undermine the skill and power basis of the Linotype operators, the highest paid of all the craftsmen. All the operators would be reduced to novices on the new board, as the inputting would now require little

more than good typing ability. This would render typists (mainly women) and Linotype operators (men) equal competitors for the new machines. The QWERTY technique was designed with an eye to using the relatively inexpensive and abundant labor of female typists.

The history of mechanized typesetting offers another instance of clear sex bias within the design of equipment. A 19th-century rival to the Linotype was the Hattersley typesetter. Compositors hated technical systems, such as the Hattersley, that separated the tasks of composing and distribution. It had a separate mechanism for distributing type, designed for use by girls. The separation of the setting (skilled) and distribution (the unskilled job of putting the letters back in their letter box) was devised as a means of reducing overall labor costs. Compositors feared that employers would try to expand this use of cheaper, unskilled labor once it got a foothold in the composing room.

The Linotype machine, on the other hand, did not represent the destruction but merely the mechanization of the compositors' setting skills as a whole. The key aspect of this successful machine was that it eliminated distribution as a task, since letters were formed anew each time by the action of brass molded on molten metal. After the type was used, it was simply melted down ready to be reused. The compositors actually welcomed the Linotype machine because it did not depend on the employment of child labor for its success. The men's union, the London Society of Compositors, even wrote a letter to Linotype Company Ltd. congratulating them: "The Linotype answers one of the essential conditions of trade unionism, in that it does not depend for its success on the employment of boy or girl labour." On the contrary, by cutting out the task of distribution, it stopped any possible inroads that boys and women might make into the trade. Thus the Linotype manufacturing company adopted a technology that was beneficial to union men.

Finally, there is perhaps another level on which the technology of production reflect male power. Feminists have understandably tended to underemphasize the material realities of physical power, given that women's exclusion from numerous occupations has been legitimated in biological terms. It is often still said that men are naturally stronger and therefore more suited to certain types of work. However, the construction of men as strong and technologically able and women as physically and technically incompetent is a social process. It is the result of different childhood exposure to technology, the prevalence of different role models, different forms of schooling, and the extreme sex segregation of the job market. The effect of this is an implicit bias in the design of machinery and job content toward male strength.

In composing work, the lifting and carrying of the forme is a case in point. The forme is heavy and beyond the strength of not only women but many men, particularly older men and younger apprentices. However, by defining this task as one that requires muscle, men seek to ensure that women workers cannot threaten to undercut men's labor. The size and weight of the forme is, in fact, arbitrary. Printing presses and the printed sheet could have been smaller, too. It is purely a question of custom at what weight the use of hoists and trolleys to transport the forme is introduced. There is nothing natural about units of work. Whether it is hay bales or 50 kg bags of cement or plaster, they are political in their design. Employers and workers have a political interest in the design of work processes. Employers prefer workers to use their brawn when it leads to more efficiency and lower production costs. Male workers use their bodily and technical effectivity to design machinery and work tasks so as to constitute themselves as the capable workers and women as inadequate.

It is overwhelmingly males who design technological process and industrial machinery. It is the knowledge and experience of engineers and of the workers who use the machines which filters through into the shape of new technologies. Mechanical equipment is often manufactured and assembled in ways that make it too big and heavy for the "average" woman to use. This need not be a conscious process or conspiracy but, rather, the outcome of a preexisting pattern of power. This is not to imply that men always design technology for their own use and in their own interests. It is more complex than that. Capital's interest cannot be supposed always to coincide with that of men as a sex. As we have seen, some technologies are designed for use by women in order to break the craft control of men. Thus gender divisions are commonly exploited in the power struggles between capital and labor. In this way, the social relations that shape technology include those of gender as well as class.

CONCLUSION

This article has proposed that skills are not technically but socially determined and that the skills that women use in their job are not fully recognized and evaluated. Skill is defined to give priority to traditionally "male" work. This has played a key role in the ability of craft unions to protect the interests of their members through restricting access to certain occupations. However, the securing of technical skills has also been a powerful

weapon in patriarchal struggles at work, excluding women from these technical skills. Finally, I have argued that male domination over tools and work relations affects not only the evaluation of women's and men's work but is often institutionalized within workplace technology. If social relations are expressed in and shape technologies themselves, then the direction of technological development must, in part, reflect existing gender relations.

The feminist analysis of gender bias in evaluating skill has been central to the campaign for equal pay. The key questions addressed to date are whether women are primarily disadvantaged by their confinement to low-skilled jobs or by the low value placed on women's jobs irrespective of their skill level. On this question hangs the critical policy question of the relative importance to be attached to equal employment policies over equal value or "comparable worth" legislation for improving women's position in the labor market. My argument here has been that we need to go beyond this to challenge the way in which the design and use of technology itself locks women into an inferior position in the labor market.

NOTES

1. In her research on the American printing industry in the late 19th century, Baran (1987) made a similar argument about publishers' attempts to transform printing into women's work following the introduction of the Linotype system.

2. Coyle's (1982) study of the clothing industry in Britain reached similar conclusions.

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